



PAN3029 470-510MHz

指标测试报告

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上海磐启微电子有限公司

测试报告

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1 Overview

This report is based on the test data of the PAN3029 evaluation board PAN3029EVB, and is intended to help customers understand the index parameters of the PAN3029 evaluation board. Since the switch combiner is added to the evaluation board, there will be a loss of about 0.3dB in power and sensitivity. At the same time, the test methods of some parameters of the evaluation board are different from the product manual, so the test results of some parameters will be slightly different from the index parameters in the product manual.

References and links:

- "PAN3029 Product Manual"
- "PAN3029 Evaluation Board User Manual"
- <https://bbs.panchip.com/forum.php?mod=viewthread&tid=8256&extra=page%3D1>

2 Test Summary

2.1 Test Content

The test contents mainly include the following aspects:

- Receiving performance test
- Transmitting performance test
- Power consumption test
- Distance test

2.2 Test Object

The entire test is based on the PAN3029EVB evaluation board, mainly testing 3 chips, and the number of tests for different projects will vary.

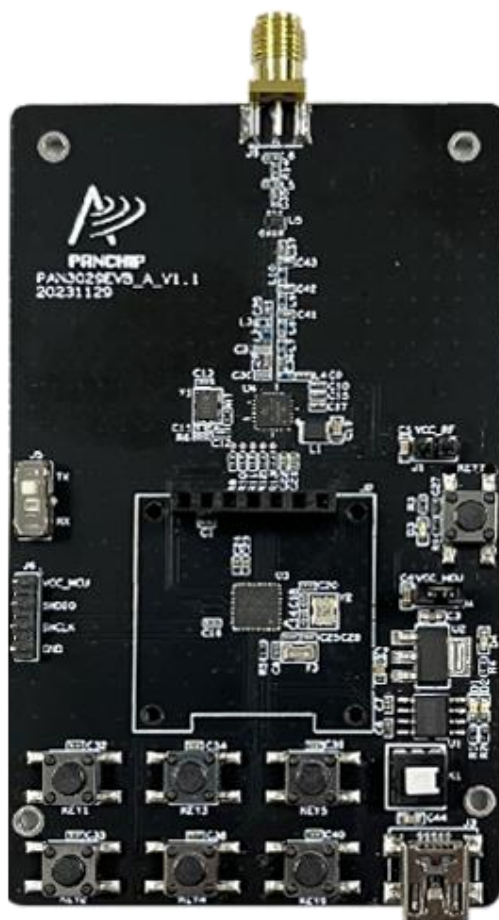


Figure 2-1 PAN3029EVB evaluation board

2.3 Test Equipment

Table 2-1 Test equipment

No.	Device Name	Device Model	Quantity
1	Spectrum analyzer	N9020B	1
2	Signal generator	N5182B/E4438C	2
3	Digital multimeter	34465A	1
4	Network analyzer	ZVB8	1
5	Shielded box	GX-5935A	1
6	USB to serial cable	/	1
7	RF coaxial cable and connector	/	1
8	Antenna	470-510MHz	2

3 Test Content

3.1 Receiving Performance Test

3.1.1 Receiving Sensitivity Test

3.1.1.1 Test Environment



Figure 3-1 Sensitivity test environment

3.1.1.2 Test Steps

1. Set up the environment according to Figure 3-1;
2. Use the button to make PAN3029 enter the LDO mode receiving state;
3. Use the button to set the parameters BW/SF/frequency/Payload Length;
4. Adjust the output power of the signal source and record the power that meets the PER requirement.

3.1.1.3 Test Results

Sensitivity of Different Modes at Different Packet Lengths

The test results are shown in the following tables:

Table 3-1 Receiver sensitivity data (PER=1%) with switch insertion loss compensated

CodingRate	Payload Length(Bytes)	BW(KHz)	SF	Sensitivity(dBm)		
				Chip #1	Chip #2	Chip #3
CR1	64	125	5	-118	-117	-118
			6	-120	-120	-120
			7	-124	-123	-124
			8	-126	-126	-127
			9	-129	-129	-129
			10	-132	-132	-132
			11	-135	-134	-135
			12	-137	-137	-137
		62.5	12	-141	-140	-141

Table 3-2 Receiver sensitivity data (PER=5%) with switch insertion loss compensated

CodingRate	Payload Length(Bytes)	BW(KHz)	SF	Sensitivity(dBm)		
				Chip #1	Chip #2	Chip #3
CR1	64	125	5	-118	-118	-118
			6	-120	-121	-121
			7	-124	-124	-125
			8	-127	-127	-128
			9	-130	-130	-130
			10	-133	-132	-133
			11	-135	-135	-136
			12	-137	-138	-138
		62.5	12	-141	-141	-141

Sensitivity of Different Frequency Points

The test results are shown in the following table:

Table 3-3 Receiving sensitivity data at different frequencies (PER=5%) with switch insertion loss compensated

Frequency (MHz)	CodingRate	Payload Length(Bytes)	BW(KHz)	SF	Sensitivity(dBm)		
					Chip #1	Chip #2	Chip #3
470	CR1	64	125	9	-129	-129	-130
478					-130	-130	-130
486					-129	-130	-129
494					-129	-129	-130
502					-129	-129	-130
510					-129	-129	-129

3.1.2 Receiving Current Test

3.1.2.1 Test Environment



Figure 3-2 Receive current test environment

3.1.2.2 Test Steps

1. Set up the environment according to Figure 3-2;
2. Use the buttons to make PAN3029 enter LDO mode & DCDC mode;
3. Use the buttons to set the parameters BW/SF/frequency point;
4. Record the RX current of different frequencies and different modes with the digital multimeter.

3.1.2.3 Test Results

The test results are shown in the following table:

Table 3-4 Receiving current at different frequencies (LDO mode & DCDC mode)

Frequency (MHz)	BW(KHz)	SF	LDO mode RX current (mA)			DCDC mode RX current (mA)		
			Chip #1	Chip #2	Chip #3	Chip #1	Chip #2	Chip #3
470	125	9	7.98	7.82	7.86	4.31	4.4	4.34
478			8	7.84	7.88	4.32	4.42	4.35
486			8.02	7.86	7.89	4.33	4.43	4.36
494			8.04	7.87	7.91	4.34	4.44	4.37
502			8.06	7.91	7.94	4.35	4.45	4.38
510			8.09	7.93	7.97	4.36	4.46	4.39

3.2 Transmitting Performance Test

3.2.1 Transmitting Power and Current Test

3.2.1.1 Test Environment

As shown in Figure 3-3.

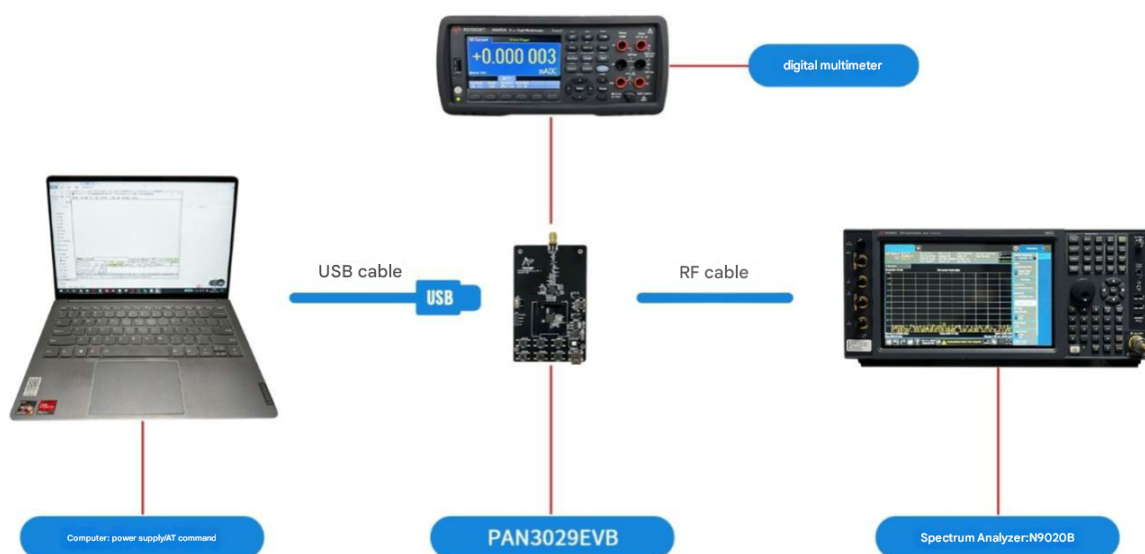


Figure 3-3 Transmit power and current test environment

3.2.1.2 Test Steps

1. Set up the environment according to Figure 3-3;
2. Use the button to make PAN3029 enter the carrier transmission mode;
3. Use the button to set the power of PAN3029;
4. The spectrum analyzer records the transmission power, and the digital multimeter records the transmission current.

3.2.1.3 Test Results

The test results are shown in the following table:

Transmitting Power and Current at Each Level

Table 3-5 Transmitting power and current at each level

Frequency (MHz)	Level	Chip #3	
		Power (dBm)	Current (mA)
490	23	19.90	97.55
	22	19.54	94.24
	21	18.34	80.84
	20	17.47	67.83
	19	16.33	60.04
	18	15.34	53.77
	17	13.71	45.52
	16	13.52	44.63
	15	13.01	42.77
	14	12.31	40.28
	13	11.20	36.76
	12	10.48	34.70
	11	7.92	26.82
	10	7.38	25.76
	9	4.51	20.50
	8	4.10	19.89
	7	2.40	17.73
	6	-1.08	14.65
	5	-1.51	14.05
	4	-2.33	13.45
	3	-5.80	12.14
	2	-17.65	8.02
	1	-26.67	7.25

Transmitting Power and Current at Different Frequency Points

Table 3-6 Transmitting power and current at different frequencies

Frequency (MHz)	Level	Chip #1		Chip #2		Chip #3	
		Power (dBm)	Current (mA)	Power (dBm)	Current (mA)	Power (dBm)	Current (mA)
470	23	20.8	117	19.7	97	20.25	108
	20	18.6	83.1	16.58	64	17.8	73
	15	14.87	55.8	12	39.3	14	49.7
	12	11.81	42.9	9.75	33.5	11.05	39.5
478	23	20.7	109.8	19.83	96.9	20.3	104.7
	20	18.9	82.85	17	63.7	18.17	74.4
	15	15.1	55.3	12.4	40	14.3	50.1
	12	11.8	41.1	10.12	33.6	11.3	38.9
486	23	20.2	99.4	19.7	92	20.04	97.3
	20	18.7	79	17.2	63.1	18.2	72.4
	15	15	52.5	12.7	39.7	14.4	48.7
	12	11.42	38.1	10.15	32.7	11.1	36.9

494	23	19.8	92.8	19.43	87.5	19.7	91.2
	20	18.57	75.2	17.2	61.7	18.05	69.3
	15	14.75	50	12.7	38.9	14.26	46.6
	12	11.1	36	10	31.6	11.27	38.9
502	23	19.56	89.9	19.34	86.8	19.57	89.8
	20	18.43	73.6	17.17	61.6	18	68.7
	15	14.6	48.9	12.7	39	14.16	46.1
	12	10.9	35.2	9.96	31.6	10.71	34.7
510	23	19.33	87.1	19.36	87.7	19.5	90
	20	17.34	62.4	17.22	63	18	69.8
	15	14.5	48.3	12.65	39.5	14.2	46.8
	12	10.67	34.5	10	31.9	10.7	35

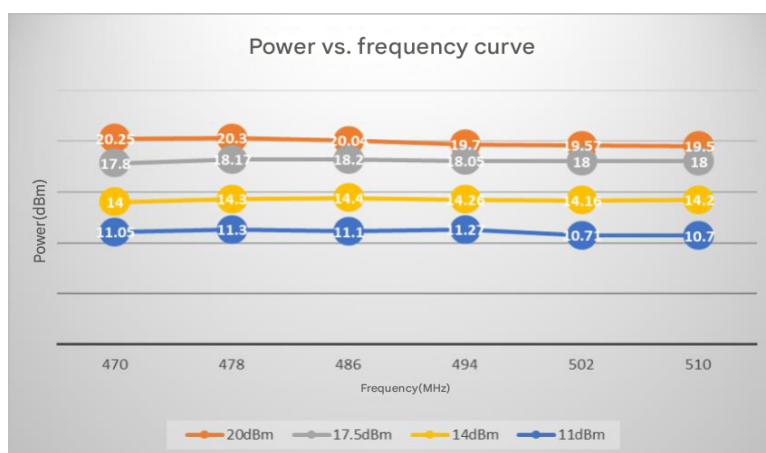


Figure 3-4 Power changes with frequency at several power levels

3.2.2 Transmitting Harmonic Test

3.2.2.1 Test Environment

As shown in Figure 3-3.

3.2.2.2 Test steps

1. Set up the environment according to Figure 3-3;
2. Use the button to make PAN3029 enter the carrier transmission mode;
3. Use the button to set the power and frequency of PAN3029;
4. Record the power and harmonics on the spectrum analyzer.

3.2.2.3 Test results

The test results are shown in the following table:

Table 3-7 Power of second harmonic and third harmonic at maximum transmission power at different frequencies

Frequency (MHz)	Level	Chip #1			Chip #2			Chip #3		
		Power (dBm)	Second Harmonic (dBm)	Third harmonic (dBm)	Power (dBm)	Second Harmonic (dBm)	Third harmonic (dBm)	Power (dBm)	Second Harmonic (dBm)	Third harmonic (dBm)
470	23	20.8	-41	-41	19.7	-43	-44.2	20.25	-41.5	-48
478	23	20.7	-42.7	-43.2	19.83	-43	-46.2	20.3	-42.8	-48.3
486	23	20.2	-45	-45.6	19.7	-44.5	-47.5	20.04	-44	-48.4
494	23	19.8	-44	-47.6	19.43	-44	-49	19.7	-43.7	-48.5
502	23	19.56	-43	-48.6	19.34	-43.7	-50	19.57	-43	-48.5
510	23	19.33	-43	-48.9	19.36	-43	-49	19.5	-42.5	-48.6

The test screenshot is shown in Figure 3-5.

Mark1: Carrier frequency signal

Mark2: Second harmonic

Mark3: Third harmonic

Mark4: Fourth harmonic

Mark5: Fifth harmonic

Mark6: Sixth harmonic

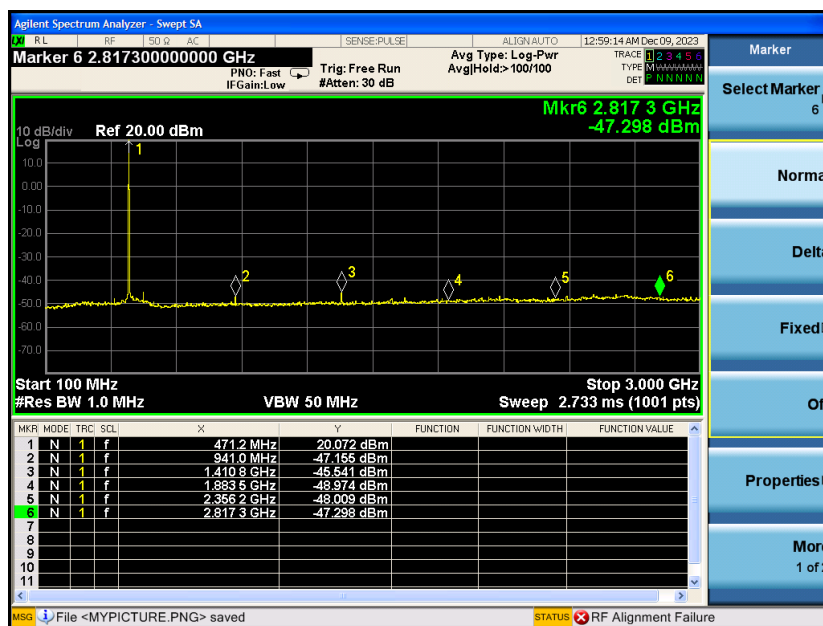


Figure 3-5 Emitted harmonics

3.2.3 Transmit bandwidth test

3.2.3.1 Test environment

As shown in Figure 3-3.

3.2.3.2 Test steps

1. Set up the environment according to Figure 3-3;
2. Use the button to make PAN3029 enter the normal transmission mode;
3. Use the button to set the BW/SF/frequency of PAN3029;
4. Record the broadband on the spectrum analyzer.

3.2.3.3 Test results

The test results are as follows:

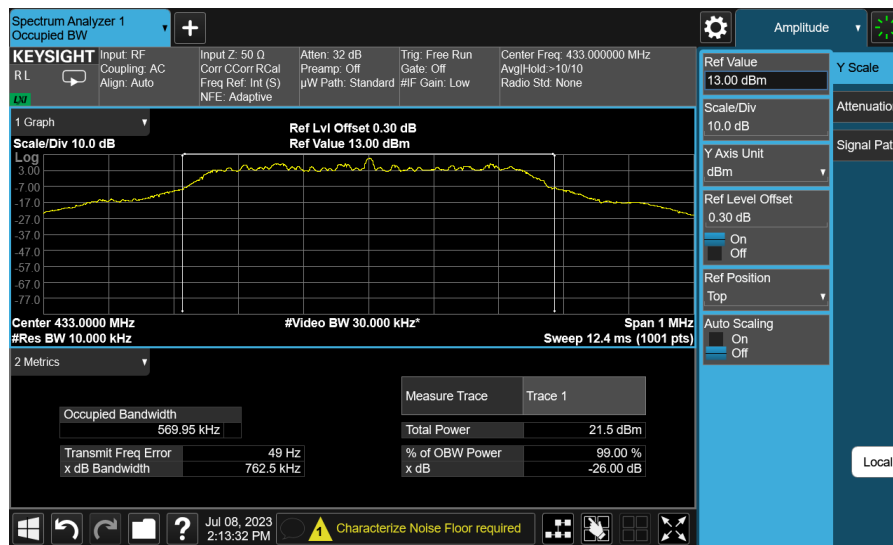


Figure 3-6 BW= 500K bandwidth

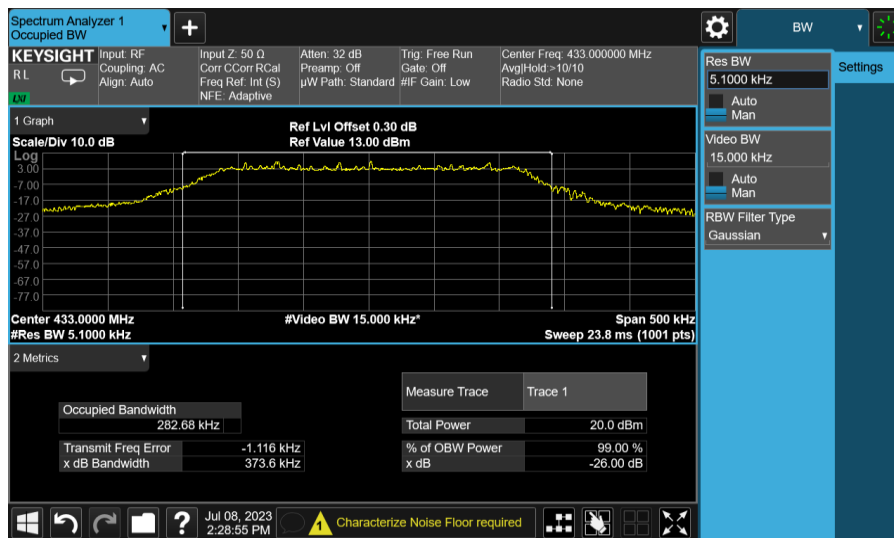


Figure 3-7 BW= 250K bandwidth

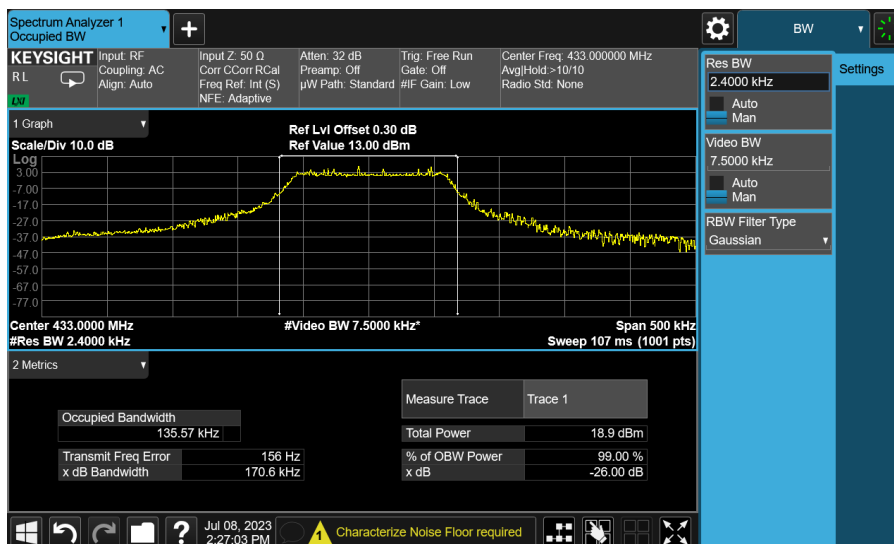


Figure 3-8 BW= 125K bandwidth

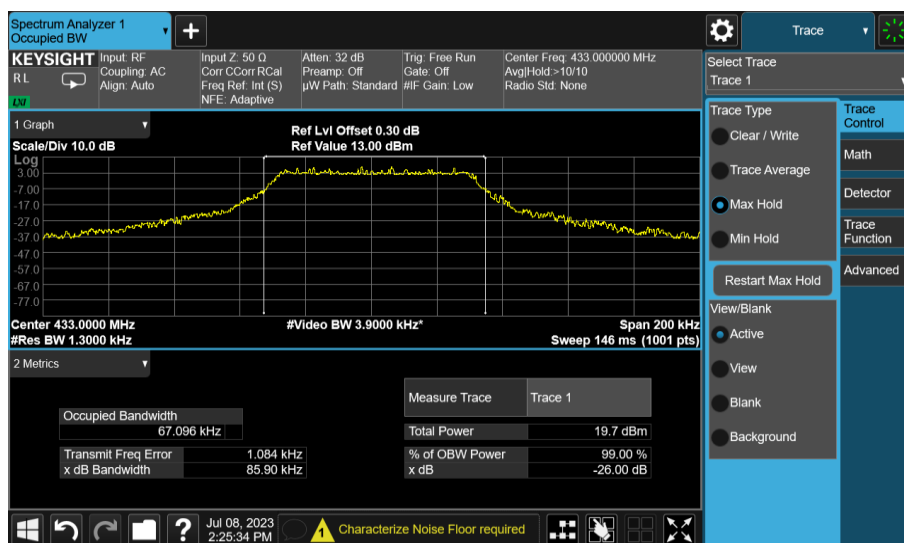


Figure 3-9 BW= 62.5K bandwidth

3.3 Low power performance test

3.3.1 DeepSleep&Sleep current test

3.3.1.1 Test environment

As shown in Figure 3-10.

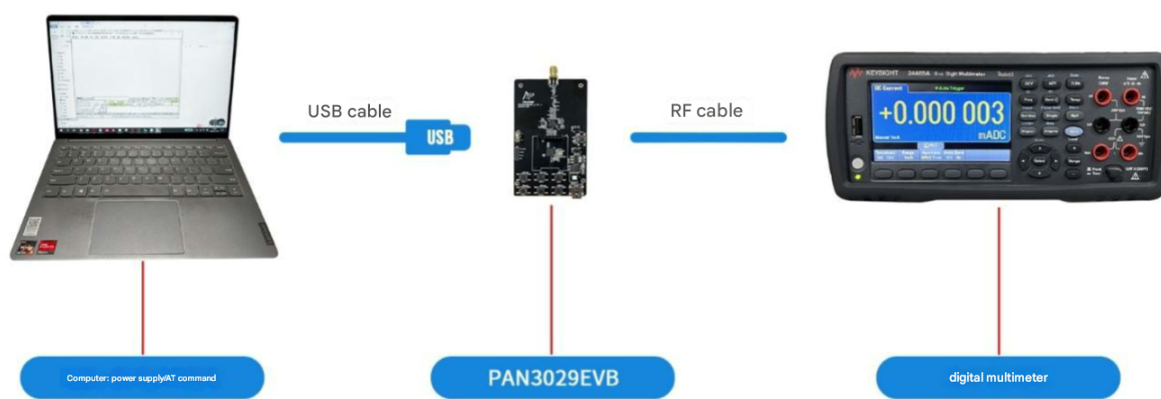


Figure 3-10 Low power consumption current test environment

3.3.1.2 Test steps

1. Set up the environment according to Figure 3-10;
2. Use the button to make PAN3029 enter the low power performance mode (DeepSleep&Sleep);
3. Record the current of the digital multimeter.

3.3.1.3 Test results

The test results are shown in the following table:

Table 3-8 DeepSleep and Sleep current

Chip	DeepSleep current (nA)	Sleep current (nA)
#1	260	621
#2	280	643
#3	245	619

3.4 Distance test in various environments

A total of four scenarios were tested, including ground-to-ground test in the urban area, rooftop-to-ground test in the urban area, lakeside-to-lake test in the suburbs, and indoor wall-to-wall test.

3.4.1 Ground communication in Suzhou city

3.4.1.1 Test environment

On Chuangyuan Road in Suzhou City, there are fewer obstacles in the urban area, more traffic lights, and more motor vehicles on the road. Map screenshots of different distances will be provided later.



Figure 3-11 Suzhou ground pull distance test environment a

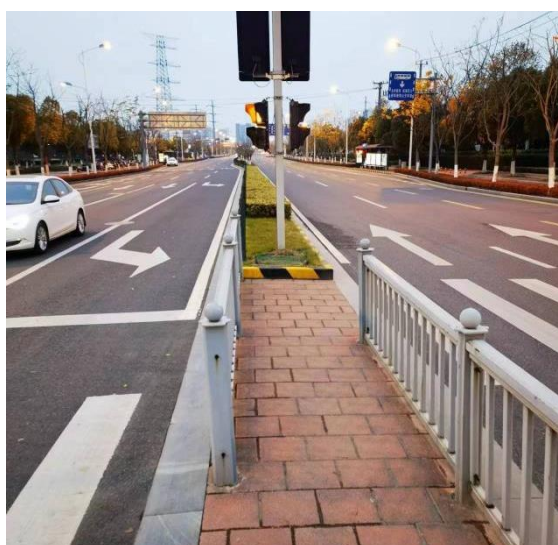


Figure 3-12 Suzhou ground pull distance test environment b

1. Module status:

- 1) Transmitter: placed in hand, 1.3m above the ground

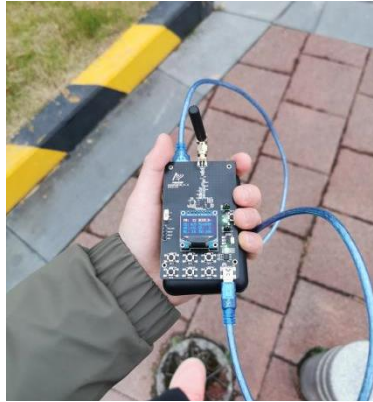


Figure 3-13 Suzhou ground distance TX end

- 2) Receiver: placed in hand, 1.3m above the ground



Figure 3-14 Suzhou ground distance RX end

2. Parameter configuration:

- 1) Frequency: 499MHz
- 2) Payload length: 10Bytes
- 3) SF and BW: There will be some changes at different distances, see test data for details
- 4) CR: 1
- 5) Output power: 20dBm

3. Screenshots of maps at different distances:

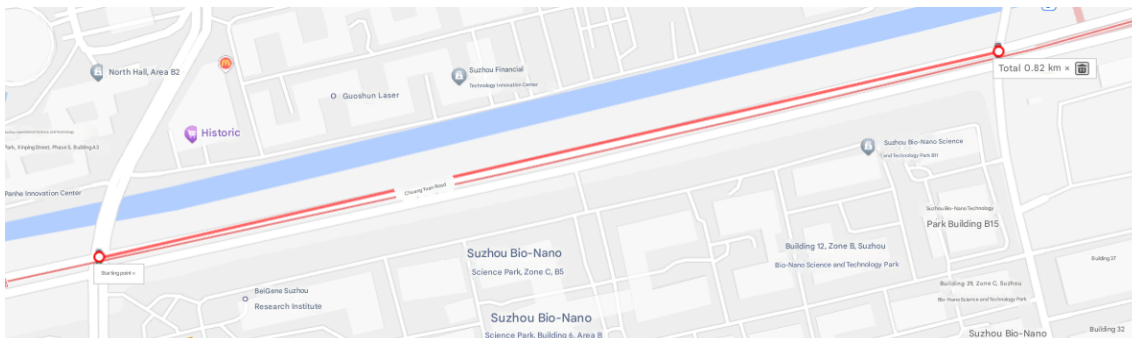


Figure 3-15 Screenshot at 820 meters

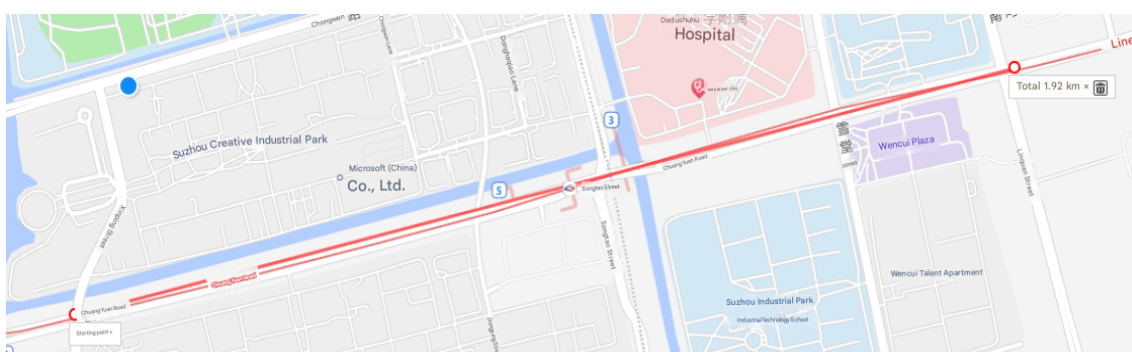


Figure 3-16 Screenshot at 1920 meters

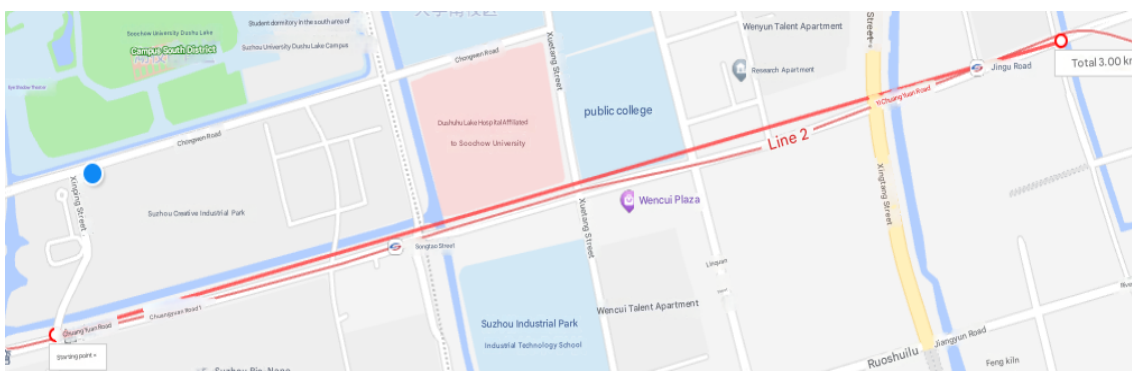


Figure 3-17 Screenshot at 3000 meters

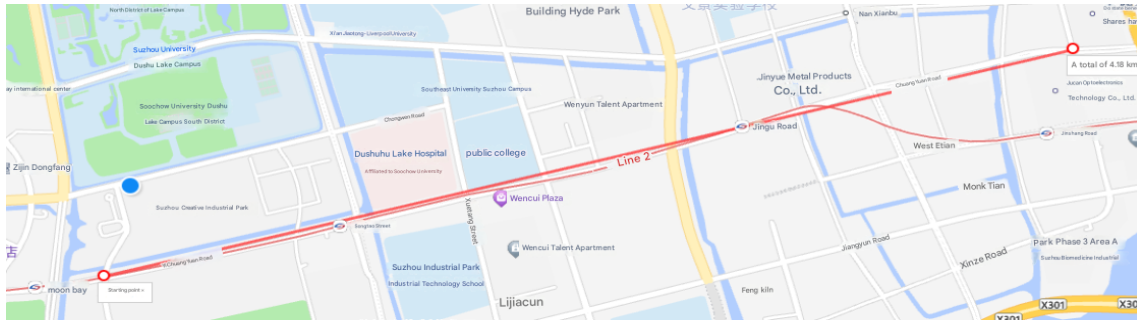


Figure 3-18 Screenshot at 4180 meters



Figure 3-19 Screenshot at 5130 meters

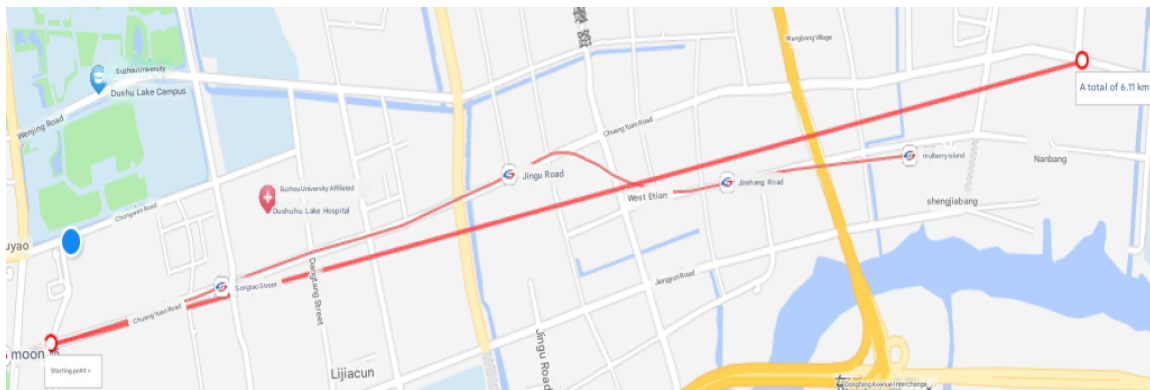


Figure 3-20 Screenshot at 6110 meters

3.4.1.2 Test steps

1. Set the TX mode and RX mode respectively through the buttons, and configure the corresponding CR/BW/SF parameters;
2. The TX end sends 100 packets of data each time through the B mode, and the RX end counts the number of packets received at different distances and different parameters.

3.4.1.3 Test results

The test results are shown in the table below. The road limit distance is 6110m, SF 12 can receive packets; 125K SF 9, 5130m can receive packets. The distance is above 4180, and the directionality is very sensitive.

The data in the table are the data with better packet reception.

Table 3-9 Ground pull distance results of Suzhou Chongwen Road

BW (kHz)	SF	Tx end position	Rx end position	Distance (m)	Packet receiving statistics	
					PAN3029	
125	5	Intersection of Chuangyuan Road and Xiping Street	Intersection of Chuangyuan Road and Dongping Street	820	94	100
	6				99	99
	7				98	100
125	5		Intersection of Chuangyuan Road and Linqun Street	1920	17	43
	6				73	96
	7				91	83
125	7		Intersection of Chuangyuan Road and Jinhai Road	3000	0	/
	8				23	55
	9				85	62
	10				90	95
125	9		Intersection of Chuangyuan Road and Jinshang Road	4180	67	88
	10				91	70
	11				80	85
125	9		Intersection of Chuangyuan Road and Guangxian Street	5130	47	20
	10				59	12
	11				74	37
	12				88	64
125	12		Intersection of Chuangyuan Road and Changyang South Street (1.3m above ground level)	6110	0	0
125	12		Intersection of Chuangyuan Road and Changyang South Street (2.0m above ground level)	6110	40	18
62.5	12		Intersection of Chuangyuan Road and Changyang South Street	6110	57	20

3.4.2 Communication between rooftop and ground in Suzhou city

3.4.2.1 Test environment

The transmitter is located on the rooftop of the 6th floor of Fuhua Technology Building, No. 199 Chongwen Road, Suzhou City. The receiver is located far away along Chuangyuan Road. The environment is relatively complex and there are high-rise buildings blocking the way.



Figure 3-21 Suzhou Fuhua Technology Building rooftop environment



Figure 3-22 Suzhou road test environment

1. Module status:

- 1) Transmitter: placed on the rooftop wall, 20m above the ground

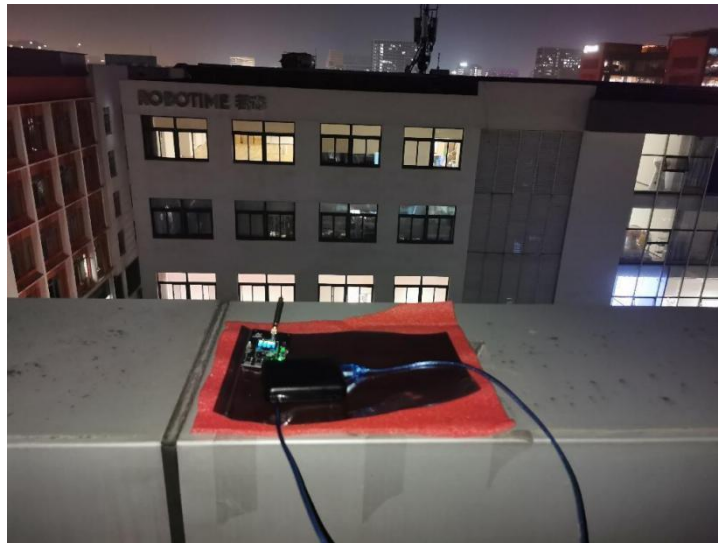


Figure 3-23 Suzhou Fuhua Technology Building rooftop TX end

2) Receiver: held in hand, 1.3m above the ground

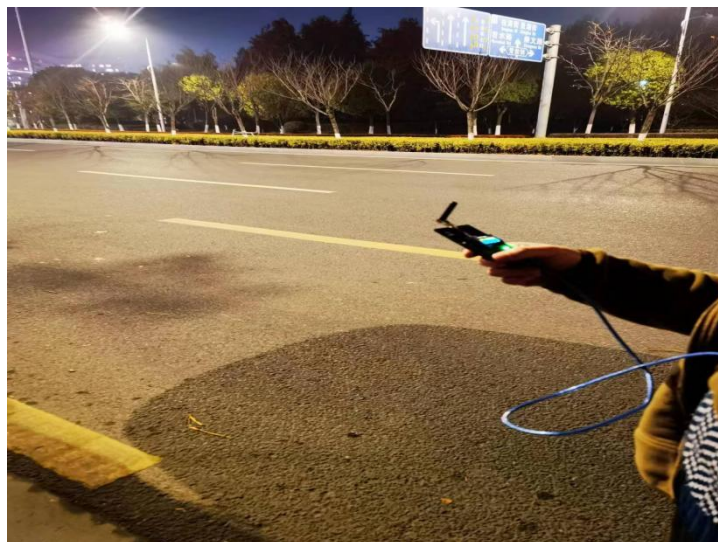


Figure 3-24 Road RX end

2. Parameter configuration:

- 1) Frequency: 499MHz
- 2) Payload length: 10Bytes
- 3) SF and BW: There will be some changes at different distances, see the test data for details
- 4) CR: 1
- 5) Output power: 20dBm

3. Screenshots at different distances:

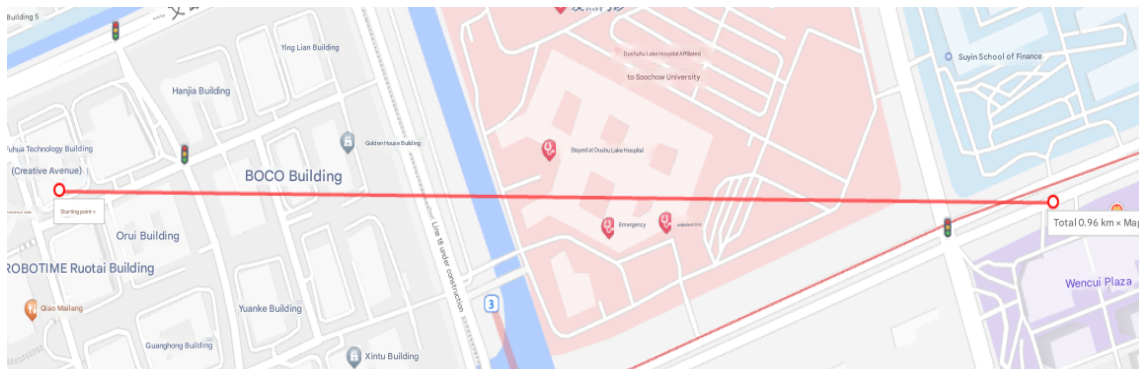


Figure 3-25 Screenshot at 960m

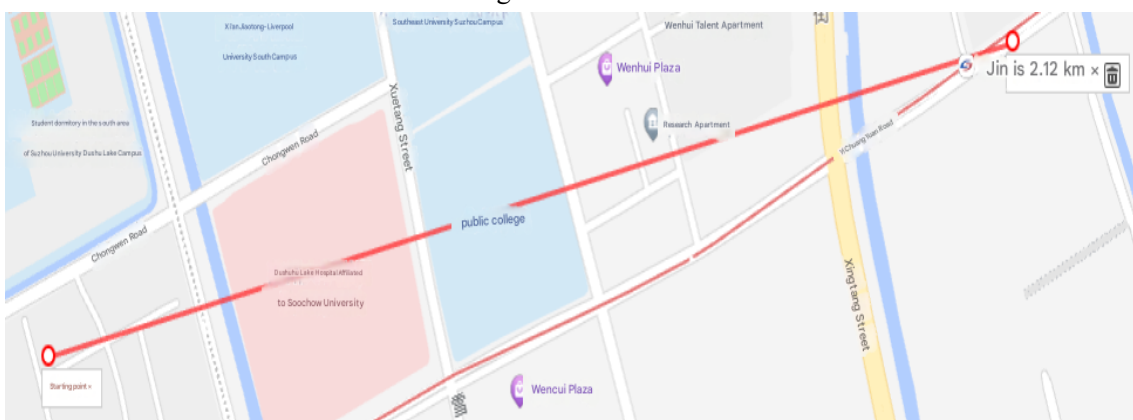


Figure 3-26 Screenshot at 2120m



Figure 3-27 Screenshot at 3510m

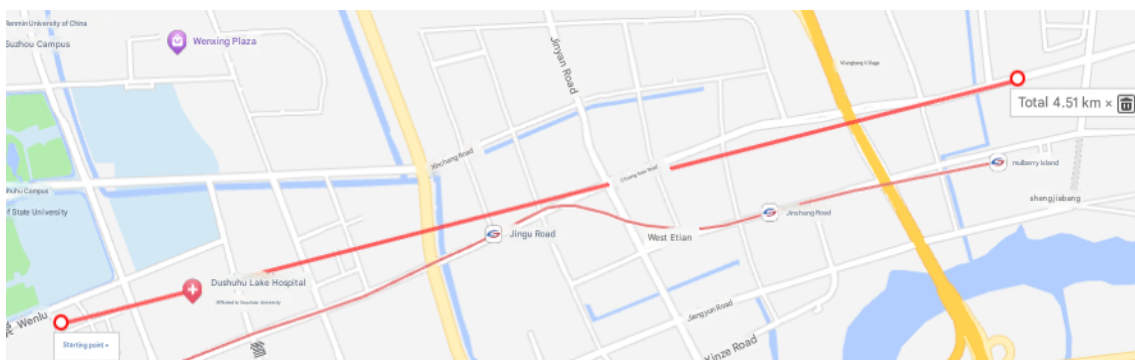


Figure 3-28 Screenshot at 4510m

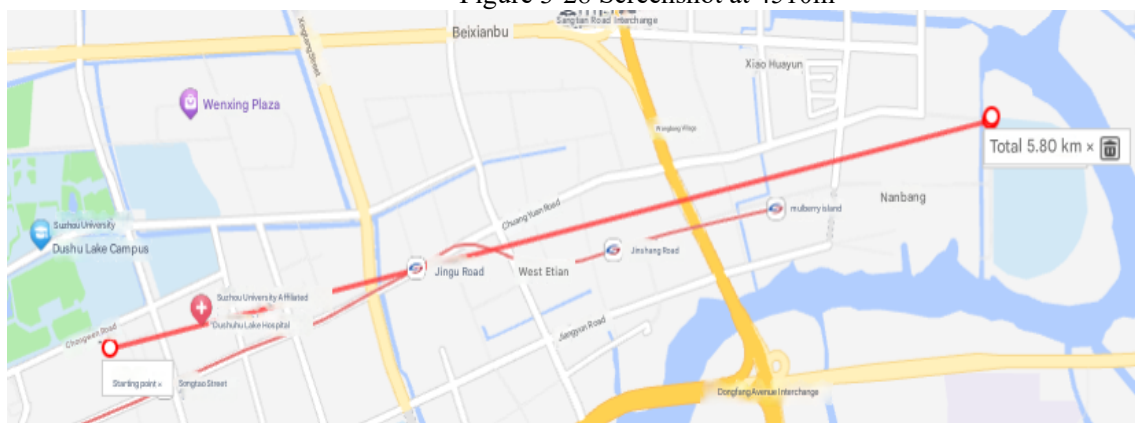


Figure 3-29 Screenshot at 5800 meters



Figure 3-30 Screenshot at 7820 meters



Figure 3-31 Screenshot at 9020 meters



Figure 3-32 Screenshot at 10000 meters

3.4.2.2 Test steps

1. Set the TX mode and RX mode respectively through the buttons, and configure the corresponding CR/BW/SF parameters;
2. The TX end sends 100 packets of data each time through B mode, and the RX end counts the number of received packets at different distances and different parameters.

3.4.2.3 Test results

The test results are shown in the table below. The extreme distance is 10km and the data packet can be received occasionally. Within 5.8km, the 250K SF12 packet reception rate reaches more than 70%. Above 5.8km, the directionality is more sensitive.

Table 3-10 Suzhou air-to-ground distance pull results

BW (kHz)	SF	Tx end position	Rx end position	Distance (m)	Packet receiving statistics	
					PAN3029	
500	5	Rooftop of Fuhua Technology Building	100m east of the intersection of Chuangyuan Road and Xuetang Street	960	39	30
250	5				99	99
125	5				100	99
	7				100	100
125	5		100m east of the intersection of Chuangyuan Road and Jingu Road	2120	4	2
	6				100	99
	7				100	99
125	7		100m east of the intersection of Chuangyuan Road and Jinshang Road	3510	25	8
	8				98	81
	9				93	98
125	8		100m east of the intersection of Chuangyuan Road and Guangxian Street	4510	0	0
	9				76	82
	10				100	100
125	9		Intersection of Chuangyuan Road	5800	23	0

	10		and Jiangyun Road		35	1
	11				16	/
	12				71	89
125	12		Intersection of Dongfang Avenue and Gangling Road (2.0m above ground level)	7820	27	42
62.5	12				21	30
125	12		Intersection of Dongfang Avenue and Xiwugu Road (2.0m above ground level)	9020	8	/
62.5	12				2	/
62.5	12		Intersection of Qingsha Road and Mingshi Road (2.0m above ground level)	10000	1	/

Note: “/” indicates that no test was conducted. All heights not marked in the table are 1.3m above the ground.

3.4.3 Communication by Taihu Lake in Suzhou suburbs

3.4.3.1 Test environment

Near Taihu Lake in Wujiang, Suzhou, with virtually no obstruction.

1. Module Status:

The test board here is an early evaluation board, not the PAN3029EVB evaluation board provided for customer samples. Customer-provided evaluation boards will perform better than this one.

1) Transmitter: Near Taihu Lake, 0.5m above ground level, slightly obstructed by reeds in front.



Figure 3-30 TX terminal

2) Receiver: Taihu Lake. Due to different environments, taking into account the convenience of testing and minimizing the impact of obstructions, the placement height will be different. There are three placement heights: 0.5m, 1.3m and 2.2m above the ground.



Figure 3-31 RX end (0.5m, 1.3m, 2.2m from left to right)

2. 参数配置:

- 1) 频点: 499MHz
- 2) Payload 长度: 10Bytes
- 3) SF 和 BW: 不同距离会有一些变化, 具体见测试数据
- 4) CR: 1
- 5) 输出功率 20dBm

3. 不同距离截图:

场景一: 发射端和接收端在太湖的同一侧, 距离 3810 米, 有点阻挡, 相对其它位置阻挡多些。

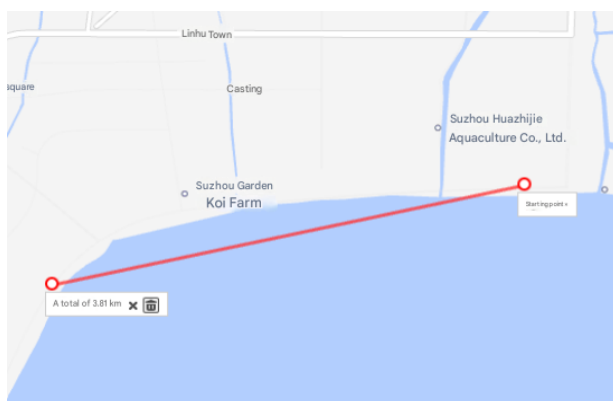


图 3-32 距离 3810 米截图



图 3-33 距离 3810 米接收端环境图

场景二：发射端和接收端在太湖的同一侧，距离 5090 米，有点阻挡，但阻挡较少。

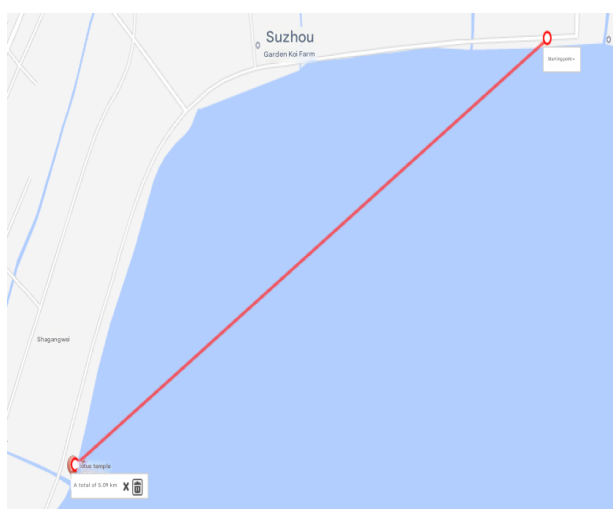


图 3-34 距离 5090 米截图



图 3-35 距离 5090 米接收端环境图

场景三：发射端和接收端在太湖的同一侧，距离 7130 米，有点阻挡，但阻挡较少。



图 3-36 距离 7130 米截图



图 3-40 距离 7130 米接收端环境图

场景四：发射端和接收端在太湖的同一侧，距离 9060 米，有点阻挡，但阻挡较少。

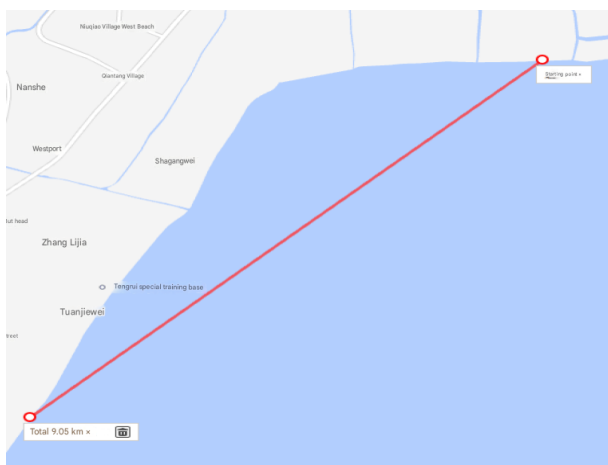


图 3-37 距离 9060 米截图



图 3-38 距离 9060 米接收端环境图

场景五：发射端和接收端在太湖的同一侧，距离 10520 米，有点阻挡，但阻挡较少。

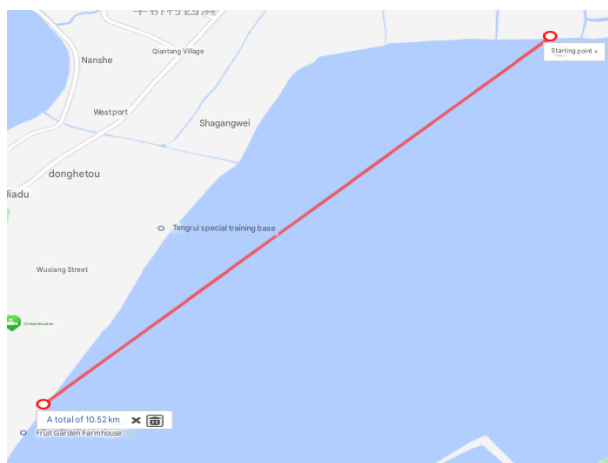


图 3-43 距离 10520 米截图



图 3-44 距离 10520 米接收端环境图

场景六：发射端和接收端在太湖的同一侧，距离 12880 米，有点阻挡，但阻挡较少。

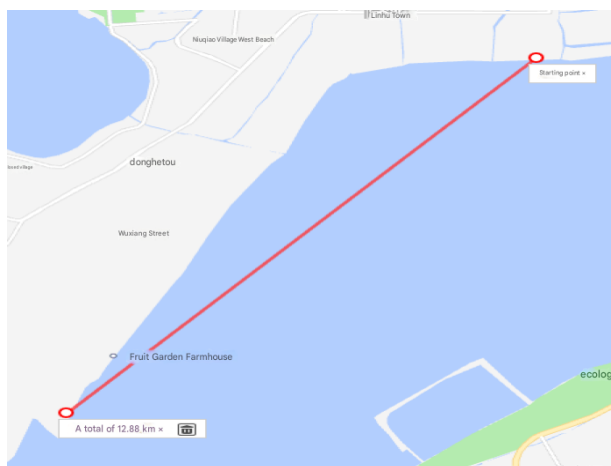


图 3-45 距离 12880 米截图



图 3-46 距离 12880 米接收端环境图

场景七：发射端和接收端在太湖的两侧，距离 14940 米，基本无阻挡。



图 3-47 距离 14940 米截图

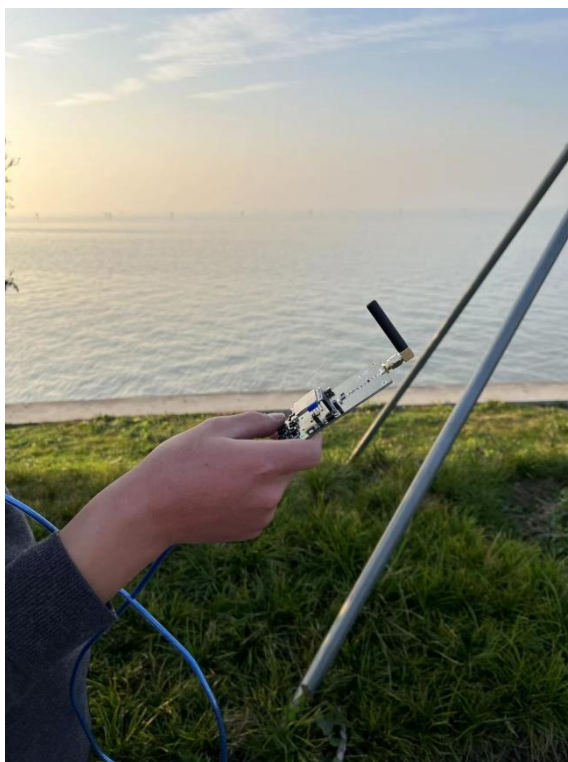


图 3-48 距离 14940 米接收端环境图

场景八：发射端和接收端在太湖的两侧，距离 17420 米，基本无阻挡。



图 3-49 距离 17420 米截图



图 3-50 距离 17420 米接收端环境图

场景九：发射端和接收端在太湖的两侧，距离 20230 米，基本无阻挡。



图 3-51 距离 20230 米截图



图 3-52 距离 20230 米接收端环境图

场景十：发射端和接收端在太湖的两侧，距离 23100 米，基本无阻挡。



图 3-53 距离 23100 米截图



图 3-54 距离 23100 米接收端环境图

3.4.3.2 Test steps

1. 分别通过按键设置成 TX 模式和 RX 模式，配置成对应的 CR/BW/SF 参数；
2. TX 端通过 B 模式，每次发 100 包的数据包，RX 端统计不同距离下和不同参数下的收数。

3.4.3.3 Test results

测试结果见下表，在 3810m 和 5090m 阻挡相对较多，125K SF10 可以收到达到 80%以上；7130m 和 9160m 收包效果很好，125K SF 10 可以收到接近满包；10520m 以上，开始出现方向性，在 17420m 和 20230m，方向性相对小一些，收包效果较好。最远到 23100m 都能收包，再远没有继续测试。

表 3-11 苏州太湖边拉距结果

BW (kHz)	SF	Tx 端位置		Rx 端位置		距离 (m)	收包统计	
							PAN3029	
125	7	滨湖大道与东山大道交汇处 往西 200m	离地高度 0.5m	滨湖大道与腾飞路交汇处往 西 300m	离地高度 0.5m	3810	4	26
	8						48	55
	9						73	75

	10						97	93
125	7			莲花寺	离地高度 0.5m	5090	0	0
	8						2	0
	9						62	59
	10						92	83
	11						92	99
500	5						0	0
250	5						3	0
125	5			滨湖大道与蔬菜园路交汇处往西南 400m	离地高度 1.3m	7130	50	40
	6						90	57
	7						73	84
	8						99	99
	9						96	100
	10						98	100
125	5			滨湖大道与石鹤山路交汇处往西南 300m	离地高度 1.3m	9060	0	0
	6						1	1
	7						52	56
	8						97	99
	10						100	100
125	10			鸡山港桥	离地高度 1.3m	10520	67	60
	11						88	85
	12						90	92
125	9			太湖湖泊生态系统野外观测站北东山分部	离地高度 1.3m	12880	11	3
	10						49	57
	11						92	99
125	11			南太湖大道与冠洁路交汇处往西北 150m	离地高度 1.3m	14940	10	/
	12						40	/
62.5	12						63	/
125	10			西溪庙港往北 150m	离地高度 1.3m	17420	6	/
	11						27	98
	12						70	100
62.5	12						97	/
125	10			南太湖大道与万宝路交汇处往西 50m	离地高度 2.2m	20230	61	/
	11						96	/
	12						100	/
							99	/
125	12			太湖温泉度假大酒店大堂往西北 60m	离地高度 1.3m	23100	42	/
62.5	12						59	/

备注：“/”表示没有进行测试。

3.4.4 Suzhou Zhongxin Building wall penetration test

3.4.4.1 Test environment

苏州中新大厦内部，墙体是实体墙，发射端在地下一楼梯，接收端在不同的位置走动，两个楼梯口中间有 8 台电梯阻挡。

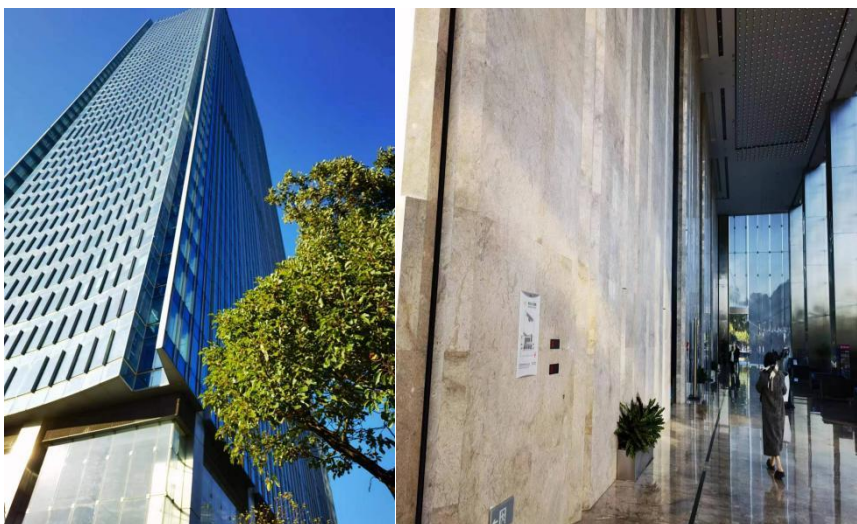


图 3-55 中新大厦

1. 模组状态:

- 1) 发射端：在地下 1 楼固定发射位置，放在台阶上，离地高度 1.4m。



图 3-396 TX 端全景图

2) 接收端：找到阻挡较多的角落，每层都放在地上，离地高度为 0.02m。



图 3-40 RX 端

2. 参数配置：

- 1) 频点：499MHz
- 2) Payload 长度：10Bytes
- 3) SF：10，BW：250KHz
- 4) CR：1
- 5) 输出功率 20dBm

3.4.4.2 Test steps

1. 分别通过按键设置成 TX 模式和 RX 模式，配置成对应的 CR/BW/SF 参数；
2. TX 端通过 B 模式，每次发 100 包的数据包，RX 端统计不同距离下和不同参数下的收数。

3.4.4.3 Test results

测试结果见下表，在中新大厦墙壁都是钢筋混凝土的建筑结构中，接收端在阻挡较大的角落内，125K SF12 可以穿墙 12 楼；收包率在 90%以上，125K SF11 和 SF12 可以穿墙 11 楼。由于穿墙测试效果跟环境强相关，具体的穿墙效果还跟环境相结合来评估。

表 3-12 苏州中新大厦内部穿墙拉距结果

BW (kHz)	SF	Tx 端位置	Rx 端位置	收包统计	
				PAN3029	
500	5	中新大厦地下 1 楼	中新大厦 4 楼	99	100

125	5		中新大厦 5 楼	99	100
250	5			99	99
500	5			61	82
125	5		中新大厦 6 楼	0	/
	6			0	/
	7			13	16
	8			56	50
	9			90	66
	10			96	97
125	5		中新大厦 7 楼	0	0
	6			33	13
	7			84	41
	8			90	76
	9			99	100
125	6		中新大厦 9 楼	2	0
	7			69	22
	8			100	100
	9			97	100
125	9		中新大厦 10 楼	0	3
	10			28	35
	11			85	94
125	9		中新大厦 11 楼	1	7
	10			70	66
	11			99	98
	12			99	99
125	9		中新大厦 12 楼	6	16
	10			66	76
	11			96	64
	12			74	52
125	12		中新大厦 13 楼	0	0
62.5	12			5	0

备注：8 层是避难层不方便对比测试，这里没有测试。

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1.0	2023.12	初版

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